Springer

Automation and Education: II. Automatizing the Classroom: Background of the Effort

Author(s): James D. Finn

Source: *Audio Visual Communication Review*, Spring, 1957, Vol. 5, No. 2 (Spring, 1957), pp. 451-467

Published by: Springer

Stable URL: https://www.jstor.org/stable/30216810

JSTOR is a not-for-profit service that helps scholars, researchers, and students discover, use, and build upon a wide range of content in a trusted digital archive. We use information technology and tools to increase productivity and facilitate new forms of scholarship. For more information about JSTOR, please contact support@jstor.org.

Your use of the JSTOR archive indicates your acceptance of the Terms & Conditions of Use, available at https://about.jstor.org/terms

Springer is collaborating with JSTOR to digitize, preserve and extend access to *Audio Visual Communication Review*

JSTOR

This content downloaded from
194.169.171.51 on Thu, 16 Oct 2025 05:35:49 UTC
All use subject to https://about.jstor.org/terms

AUTOMATION AND EDUCATION:

II. Automatizing the Classroom—Background of the Effort

• JAMES D. FINN

This is the second in a series of articles dealing with various aspects of the relationship between automation and education. The first, which appeared in the Winter, 1957, issue of the REVIEW, *dealt with the general relationships between the two. This article reviews the specific proposals for automatizing the classroom through audio-visual devices in terms of the organizations leading the movement and the theoretical justification for these proposed changes in educational procedure. James D. Finn is professor of education at the University of Southern California and educational consultant for* Teaching Tools *and* Film World *magazines.*

THE FIRST paper in this series (5) developed the background of that phase of the Second Industrial Revolution known as automation. It also attempted to explore generally the current and future relationships that exist or will exist between education as an institution and automation as both a complex of industrial processes and a concept of wide application to human affairs. This general study attempted to supply background so that a closer look might be taken at the application of the concept of automation to education itself.

As indicated in the first paper, the student of audio-visual communication has a special interest in any proposals that the educational process become "automated" or "automatized." The majority of these proposals have suggested that audio-visual devices, particularly television, are the means by which many pressing problems of education can be solved. In general, these proposals have been made by persons or agencies outside of the mainstream of the audio-visual movement as it has developed in this century. Much of the major work now being done in exploring this field is being carried on by groups that have not been closely associated with previous audio-visual developments. Floyde Brooker, execu-

This content downloaded from
194.169.171.51 on Thu, 16 Oct 2025 05:35:49 UTC
All use subject to https://about.jstor.org/terms

tive secretary of the Department of Audio-Visual Instruction, NEA, has recently pointed out that ". . . many superintendents these days are talking about educational automation (films, television, etc.) seemingly without realizing that this *is* audio-visual education" (1:64).

It will be the purpose of this paper to present to the audio-visual field a statement of the rationale surrounding the proposals that audio-visual communication devices and techniques can bring automation into the educational process and thereby solve many of the most perplexing of educational problems now and in the future. This paper will discuss educational (audio-visual) automation under two general headings: (a) the institutional background, and (b) the theoretical position.

THE INSTITUTIONAL BACKGROUND

In the decade between 1930 and 1940, two institutions can be said to have given the audio-visual movement great impetus. These institutions were the Payne Fund of New York and the Rockefeller Foundation, principally through one of its funds known as the General Education Board. The so-called Payne Fund studies on the effects of the theatrical motion picture on children and youth and the subsequent motion picture appreciation movement are now classics, not only for students in the audio-visual field, but in the field of social psychology as well.

The work of the Rockefeller Foundation is, perhaps, not as well known, but was tremendously influential. Probably the most important project created by the Rockefeller grants was the Motion Picture Project of the American Council on Education. Here, Charles Hoban, Floyde Brooker, and Francis Noel, to name but a few, discovered and perfected techniques that later almost totally governed the audio-visual programs of the Navy (Noel), the Army (Hoban) and the U. S. Office of Education (Brooker) during World War II. Those who have ascribed the startling developments in audio-visual education post-1945 to "the Armed Services" would do well to review the history of the four years immediately preceding the war when the American Council Project was in full swing.

Other projects financed by the General Education Board and the Rockefeller Foundation included the Evaluation of School Broadcasts project at Ohio State University under the direction of I. Keith Tyler, the Minnesota motion picture production project

This content downloaded from
194.169.171.51 on Thu, 16 Oct 2025 05:35:49 UTC
All use subject to https://about.jstor.org/terms

under Robert Kissack, the Wisconsin Radio Study under Lester W. Parker, the human relations film excerpting project under Alice Keliher (with an assist from the Motion Picture Producers Association), a project with recordings, a project with film laboratories which later resulted in EFLA, the Rocky Mountain Radio Council, and several more. In addition, the Foundation trained personnel by providing fellowships for qualified people—permitting them to work in these projects, in the radio networks, and even to study broadcasting in England and Europe.

This bit of history, not too well known by many people presently engaged in the audio-visual movement, is extremely important in that it established the influence of outside institutions on what was, until the time of these projects, a halting, and almost ineffectual effort to bring audio-visual devices into American education.[1] It can be said categorically that for 15 crucial years (1930-1945) the audio-visual movement was influenced critically by two institutions—the Payne Fund and the Rockefeller Foundation. One additional point should be made clear, however. The Rockefeller Foundation, in particular, had no *specific* program for American education. The influence it provided was simply that it made money available for research, development, and the education of personnel. The result of all this activity—much good, some indifferent, a little useless—was the responsibility of the people and institutions who received the money. *The result, however, was that the audio-visual field developed and grew as it never had before.*

We come now to the present. The impetus toward this new idea of educational automation via the audio-visual route is also the product of an institution. That institution is the Ford Foundation. Much of the impetus has come from the Fund for the Advancement of Education (now merged back with the parent Ford Foundation) and its able executive vice president, Alvin C. Eurich. However, the Fund for Adult Education (another subsidiary) has also contributed with its support of educational television and educational television stations. Most of the experimental programs in closed-circuit television have been sponsored by the Fund for the Advancement of Education (in some cases, as at Hagerstown, Md., in cooperation with other groups).

[1] The Carnegie Foundation's Committee on Scientific Aids to Learning should not be forgotten in this account, and the contributions of earlier philanthropic grants such as the Commonwealth study (Freeman) and the Eastman Kodak study (Wood and Freeman) have to be recognized. It is not the writer's purpose, however, to write an entire history of grants for audio-visual research and development. Rather, it is to point out the tremendous influence on the development of the movement resulting from the Rockefeller and Payne contributions.

This content downloaded from
194.169.171.51 on Thu, 16 Oct 2025 05:35:49 UTC
All use subject to https://about.jstor.org/terms

There are two differences between the foundation support of the 1930-1940 period and the present. The first has to do with the amount of money expended. Evening out differences in the value of the dollar, the Ford group obviously has invested much more money in its current audio-visual projects than did the Rockefeller group. Secondly, this money has been made available in a shorter time.

However, a still greater difference exists between the two types of foundation influence, and this difference is qualitative and not quantitative. The Ford Foundation operates on a different philosophy than most of the foundations that were created in an earlier day. The Ford Foundation develops a program and sets out to implement it with grants. In the case under consideration—automatizing the classrooms of the nation—the Ford Foundation has set this as a desirable goal and has made grants for both research and development programs in order that this goal may be reached.

This implementation of a goal with money distributed to institutions in locations as far apart as Pittsburgh and San Francisco would seem to be a more efficient way of exerting influence. While the older foundations may have agreed that an area of endeavor was worth support, groups or institutions went to them with proposals, and grants were made pretty much in terms of the proposals *per se*, not in terms of whether or not the proposal fitted a program the foundation had already set up as desirable.

It may seem, upon cursory consideration, that this excursion into foundation philosophy and procedure may be an unwarranted digression in a discussion of the audio-visual automation of education. *Actually, this is probably the most important point that could be considered.* It has been established that the earlier—should we say laissez faire—grants had a large influence on the audio-visual movement. It must be soberly realized that, in the present case, we have in operation a tremendously wealthy and influential institution with a specific program for changing educational procedures, methods, and organization in the direction of classroom automation with audio-visual communication devices. *It is entirely reasonable to predict that this influence will be greater and will make itself felt in a much shorter time than anything heretofore experienced, including World War II. In studying the concept of automation in the classroom, this is the primal fact.*

This content downloaded from
194.169.171.51 on Thu, 16 Oct 2025 05:35:49 UTC
All use subject to https://about.jstor.org/terms

The Ford Foundation, similar to other foundations, for the most part does not conduct its activities itself. Leaving out, for the moment, grants to educational institutions, school systems, and such, it has created several other institutions which now become part of the institutional background of the movement toward automatizing the classroom. It would be well, in order to round out this institutional background, to list these institutions and define the areas in which they are working.

According to the 1956 *Annual Report* of the Foundation, more than $11 million had been allocated, prior to the year covered in the report, to educational television in its various phases (6:35). During the year covered by the 1956 *Report,* an additional $8 million was added to this amount. Organizationally, the money was allocated as follows:

Educational Television and Radio Center, Ann Arbor, Michigan	
General Program	$6,263,340
Public Information	90,500
National Association of Educational Broadcasters	94,000
Joint Council on Educational Television	140,000
Committee on Television, American Council on Education	12,000
Total organizational grants	$6,599,840

(6:37-38)

The Foundation also has allocated $1.5 million from which to make grants to colleges and universities so that they can release time of faculty members for participation on educational telecasts. This grant is in the form of a matching grant, requiring each individual institution participating to partially match a grant of $37,500. The telecasts involved are apparently both closed-circuit and broadcast.

Mention should also be made of the underwriting of the Omnibus program which, although in a sense commercial television, is obviously educational. The productions of Omnibus by the Ford Foundation's TV-Radio Workshop also pay extra educational dividends. Some 10,000 requests for kinescopes have been received from schools and colleges (6:44). Films compiled from past programs are being distributed to additional television stations, and some have been placed on loan with the Educational Television and Radio Center. During the year covered by the 1956 *Report,* the TV-Radio

This content downloaded from
194.169.171.51 on Thu, 16 Oct 2025 05:35:49 UTC
All use subject to https://about.jstor.org/terms

Workshop was underwritten with $1,733,887. Sponsors furnished $1,058,017 in income, leaving a deficit of $675,870 covered by Foundation funds.

Excluding direct grants (mostly through the Fund for the Advancement of Education) to schools and colleges for closed-circuit and special developments, including grants as well as paid-out moneys for the items noted above, and evening out the TV-Radio Workshop figures, the Ford Foundation has put approximately $8,700,000 in educational television in a one-year period. Over $7 million of this amount has been placed through organizations, and these organizations now become the dominating influence of this movement. Briefly, the organizations receiving over $6.5 million carry out the following functions:

Educational Television and Radio Center

By all odds, the dominant force in the educational television movement in the United States today, the Center develops and distributes programs to the affiliated ETV stations (six hours a week, kinescope and film, to be increased to ten hours a week); it provides information on ETV to the general public and to interested organizations, consulting services, speakers, etc. Many of the series produced under its subsidies are available in film form for use as audio-visual aids or on closed-circuit installations through a distribution arrangement with the University of Indiana. The Center has a distinguished Board of Directors, a large staff under the direction of Harry K. Newburn, former president of the University of Oregon, and has a program of bringing in, for a year at a time, specialists in film-making or television in the capacity of program associates. With the possible exception of Robert B. Hudson, Program Coordinator of the Center, not one member of the Board or of the staff has had anything significant to do with the development of audio-visual education in the United States prior to the establishment of the Center.

Joint Council on Educational Television

The JCET essentially is concerned with the preservation and utilization of educational television channels. It maintains legal representation in Washington, appears before the FCC, and gives advice to communities and institutions seeking aid in applying for and constructing ETV stations. In general, the members represent the higher institutions and public school systems of the United States and the operation of the JCET is in some way related to that of the American Council on Education, although this relationship is not clear from the literature (The American Council many times serves as the voice of education, particularly higher education, in Washington). The members of the JCET are all distinguished educators, none of whom has ever been associated with the audio-visual movement.

The National Association of Educational Broadcasters

The NAEB was a relatively small association of college and university radio stations until the advent of television and foundation subsidy. Its membership now includes educational institutions that operate either TV or radio stations, institutions which produce programs for such stations, and has re-

This content downloaded from
194.169.171.51 on Thu, 16 Oct 2025 05:35:49 UTC
All use subject to https://about.jstor.org/terms

cently, by way of a merger, acquired the membership of the old Association for Education by Radio and Television (AERT). It has an imposing list of services to its members—technical and program consultation—and has a wide range of very valuable publications. The NAEB also operates a radio-tape network among educational institutions. Among the officers, names long associated with educational radio seem to be dominant. As of January, 1957, Burton Paulu, of the University of Minnesota, was listed as president, and Harry J. Skornia as executive director.

Committee on Television, American Council on Education

Compared to the three agencies above, the Committee on Television is a small operation. According to the 1956 *Annual Report* of the Ford Foundation, the Committee exists to maintain liaison with "those educators not formally engaged in the educational television movement" (6:39). Eight of the 14 members are college presidents or other high administrative officers of colleges and universities, two are public school superintendents and the other four represent agencies, associations, and like organizations. The Committee has held several conferences and produced some publications.[2]

The grants to and for these organizations and their programs have had the effect of forwarding the entire educational television effort without necessarily pinpointing the direct classroom, automation idea. The grants for such purposes have been made mainly through a subsidiary fund of the Ford Foundation, the Fund for the Advancement of Education, as noted above. The majority of these grants has been made since the 1952-1954 *Report* of the Fund (8). In that report, only one grant, $29,300, is reported as being awarded to the Montclair (New Jersey) State Teachers College to investigate the possible uses of television as a classroom aid (8:90).

Figures are not easily available since the 1952-1954 *Report* and not all accounts of the projects developed report the amount of money involved. Lombard refers to the Fund grant to San Francisco State College as ". . . over a quarter million dollars" (11:19). Carpenter and Greenhill mention that their first proposal to the Fund ". . . would cost over one hundred thousand dollars" (2:11). A search of all major libraries in the Los Angeles area revealed no report from the Fund for the Advancement of Education since the 1952-1954 *Report*. However, an approximate figure was turned up in the *NET News* for May-June 1956—a figure which can be considered reasonably accurate, as the *NET News* is published by an agency subsidized by the Foundation. The quo-

[2] The situation described above, *i.e.*, that the entire television field is dominated by agencies outside the DAVI, is a serious indictment of the officers of the DAVI (including the writer) and of the national staff. Space does not permit an analysis of the situation from the professional organization point of view and to raise questions why the Ford Foundation and its agencies have decided to ignore or shortcut the existing national educational organization which, for over 20 years, has been concerned in these matters. Such an analysis should become the number one item on the agenda for the DAVI Board of Directors.

This content downloaded from
194.169.171.51 on Thu, 16 Oct 2025 05:35:49 UTC
All use subject to https://about.jstor.org/terms

tation reads, "In the past two and a half years, the Fund has awarded 21 grants totalling $1,590,286 to six colleges, eight universities and four public school systems for experiments in closed and open circuit television teaching" (p. 3).

Accepting these figures as accurate, the total investment of the Ford Foundation and its subsidiaries in educational television in the space of less than four years would amount to over $20 million. This is a very large amount of money, indeed, and would probably amount to almost 20 times as much as was invested by the Rockefeller Foundation and the Payne Fund in the earlier decade. The influence of such an investment will, if history is any guide, be very marked. The amount already expended by the Fund for the Advancement of Education alone can be expected to exert greater influence on American educational procedures than the sum total of all the efforts of audio-visual professionals or of the DAVI since its post-war reorganization.[3]

There are, of course, compelling reasons behind this interest and financial support of educational television and of studies and projects designed to explore the possibility of automatizing the classroom. The next section will describe the theoretical position (and certain practical factors) which has guided the efforts of these organizations and groups as they attempt to reorganize, re-orient and otherwise rebuild American educational procedures in the direction of educational automation.

THE THEORETICAL POSITION FOR THE AUTOMATIZED CLASSROOM

Before outlining the theoretical position upon which all of the effort toward introducing television (and, to a lesser extent, motion pictures) into the educational process is based, it might be well to state explicitly just what is meant by automatizing the classroom. Essentially, the proposal is to introduce entire systems of audio-visual experiences into the classroom *in lieu* of the teacher. Thus, a fifth-grade arithmetic lesson may be taught to large numbers of 10-year-olds in a single school, a school system, or an entire county

[3] An interesting numbers game can be played in this connection. If you take Stoddard's figures as to current enrollments in elementary and secondary education as 38 million (15:15) and multiply them by the median amount per pupil spent on audio-visual education in the United States in 1953-1954 according to the NEA Research Division ($0.65) (13:118), the answer comes out about $24.75 million—not too much under the $20 million figure for the Ford grants. Actually, the figure developed from the NEA per-pupil factor is probably high, since the study received replies from only about one third of the school districts in the United States. The non-responding districts were mainly in the smallest categories where no audio-visual programs might be expected. It is probably pretty close to the truth to say that the Ford grants for educational television have been equal to or have exceeded the entire amount of money spent on audio-visual education in the public schools of the United States in a year.

This content downloaded from
194.169.171.51 on Thu, 16 Oct 2025 05:35:49 UTC
All use subject to https://about.jstor.org/terms

by one master teacher using closed-circuit television as the medium. In a recent project undertaken by Encyclopaedia Britannica Films in connection with the ETV station in Pittsburgh, an entire course in high school physics was filmed, using a master teacher. These films may be broadcast, sent out over closed-circuit, or projected in individual classrooms. In any case, the films (162 thirty-minute subjects) would be used to convey the content of the course. Any school district may obtain the services of this expert film teacher for approximately $13,000.

The fundamental basis for this idea is the continuing teacher shortage coupled with the tremendous increase in enrollments at all levels of education from kindergarten through college. This is basic in all statements, speeches, and articles appearing on the subject. Nowhere, I suppose, is the proposition more baldly stated than by Harry Skornia, of the NAEB, as quoted by Richard Hull:

> As NAEB Secretary Harry Skornia says, education's crisis is the firmest basis for ETV's growth. We (the NAEB) were prepared for educational television. We were able to lead the way.

Hull then goes on to expand on this point of view:

> For better or worse, the rough outlines of a national framework for radio and television education have now been sketched out in a time of population pressure and an increasing array of problems for all levels of education. . . . The question is not whether to use educational television or educational radio in the educational crisis—the question is whether or not the personnel and the facilities available will be *sufficient* to meet the need in the future. (9:34)

Other evidence may be adduced to show that the crisis in teaching is basic to all thinking on the subject. Carpenter and Greenhill, in their report of the research at Pennsylvania State University on closed-circuit television, indicate that they approached the Fund for the Advancement of Education with a proposal for what might be described as "pure" research in the field of television methodology using only segments of some courses for experimental purposes. The reply they received is described as follows:

> Officers of the Fund reacted to this proposal by stating the following points either explicitly or by implication:
>
> 1. *The Fund's principal concern was with the impending shortage of teachers.* (Italics supplied.)
>
> 2. Closed-circuit television seemed to have possibilities for making it feasible for a few good instructors to reach and influence large numbers of students. (2:11)

Carpenter and Greenhill went on to say that the Fund wanted

This content downloaded from
194.169.171.51 on Thu, 16 Oct 2025 05:35:49 UTC
All use subject to https://about.jstor.org/terms

whole courses taught by television, the feasibility of the medium studied, and the acceptability by faculty members explored.

Readers of this paper are no doubt familiar with the general outline of the facts surrounding the existing and anticipated teacher shortage. However, to construct the rationale for proposed classroom automatization, it would be well to review them briefly. Two major trends are involved: (a) the rise in enrollments at all levels of schooling, and (b) the subsequent shortage of teachers. Two publications of the Fund for the Advancement of Education (7), (15) make the enrollment trend crystal clear. Stoddard points out that there are ". . . in excess of 29 million in our elementary schools (by 1965, 36 million), over 8 million in our secondary schools (by 1965, 12 million), and over 3 million in our colleges (by 1965, 4.5 million)" (15:15-16). These startling figures are compounded by the fact that, in addition to the rise in total enrollment, the *rate* (percentage of those in any given age group) attending school is also rising. For example, 62 percent of the 14-21 age group are now in school, compared to only 8 percent in 1900. By 1960, it is expected that elementary school enrollments will be 68 percent above 1946, and, by 1969, secondary school enrollments will be 70 percent above the 1954 level (7:12).

College enrollments are somewhat harder to predict. The Ford study used two projections, one anticipating no rise in the *rate* of those attending college, the other taking a rise in rate into account. The report, however, points out that whatever projection is taken, ". . . it is evident that college enrollments are likely to reach double their 1954 level sometime between 1966 and 1971" (7:14).

The tremendous increase in enrollments obviously has a direct relationship to the number of teachers needed and the number that can be obtained. "In order to provide for replacements, expansion and the maintenance of present pupil-teacher ratios, the schools of the nation must find 16 new teachers between now and 1965 for every 10 teachers now on the job. This is the equivalent of replacing all the teachers we now have and finding 60 percent in addition—all within 10 years" (7:19). With reference to college teachers, the figures are somewhere between 16 and 25 new teachers needed by 1970 for every 10 existing today.

The matter of quality of teaching also enters at this point. Where can that many qualified teachers be obtained? And, for

This content downloaded from
194.169.171.51 on Thu, 16 Oct 2025 05:35:49 UTC
All use subject to https://about.jstor.org/terms

that matter, how may they be obtained? The Ford study states, "About one-fifth of all 1954 graduates of four-year colleges entered school teaching. But during the next ten years one-half of all college graduates . . . would have to enter school teaching. . . . Nothing approaching this proportion . . . can be expected to enter teaching" (7:23). Stoddard estimates the shortage of fully qualified teachers at 250,000 in 1965 (15:17). At the college level, where the PhD is considered (rightly or wrongly) the mark of qualification, it is estimated that the number of *qualified* college teachers will decline *50 percent* by 1970.[4]

Coupled with the teacher shortage is a shortage of buildings and plant. While not so directly concerned with the problem, a consideration of physical plant is certainly germane because the nature of the methods used to solve the teacher shortage (class size, television, etc.) would have an effect on both the amount and nature of the plant constructed. The Twentieth Century Fund, in a special study, estimates the need under present instructional conditions for new physical plant in the public schools through 1960 to be in the neighborhood of $30 billion (16:4). Less than half of that amount is now being spent. Capital needs for *public* higher institutions (including dormitories, etc.) were estimated by McDonald at about $15 billion by 1975 (12:256). As an example from one state, Semans and Holy early this year advised the California State legislature that the capital outlay needed to expand existing units of the University of California and the California State Colleges and to add new units as necessary to meet swelling enrollments would cost approximately $1 billion by 1970 (14:79, 97). Junior Colleges were not considered in this estimate.

These are, as the philosophers say, the brute facts of the situation. The word "crisis" has been overworked, but the shortage of teachers and lack of plant facing this country in the next 20 years *is* a crisis of the first magnitude. When considered in the context of the cold war, the atomic age, the demands of automation, and the expansion of knowledge, the question can well be asked whether or not a technological society such as we are attempting to build can, in fact, provide its own means for *survival,* not to mention advancement.

[4] This estimate is probably too conservative. It assumes that almost the same proportion of PhD's will enter college teaching as did in the 1930's. The technical requirements of industry and government have, however, changed radically. One of the characteristics of an age of automation is, apparently, a shortage of educated talent. The percentage of PhD's likely to enter college teaching will probably decline against the competition of industry and government.

This content downloaded from
194.169.171.51 on Thu, 16 Oct 2025 05:35:49 UTC
All use subject to https://about.jstor.org/terms

The crisis has not gone unnoticed and some efforts at solutions have been made. The picture, however, is mixed. At this writing, Federal aid for school construction is bogged down in Congressional disputes; the Federal Housing loan program for colleges and universities is in danger. Some local communities are becoming reluctant to vote more taxes for schools. Gifts by industry to higher education—much of which is in the form of scholarships—are on the increase. Many states and many communities are making valiant efforts to increase school support. On the other hand, the United States Chamber of Commerce and certain widely read columnists, such as Raymond Moley, stoutly maintain that no crisis exists and the whole thing is an invention of the NEA.

Into this scene of crisis has flowed money and counsel from the Ford Foundation and its several Funds, from organizations, committees, and conferences which have been helped or stimulated by Ford money and ideas. Leading the effort has been the Fund for the Advancement of Education, and the latest statement of the position of the Fund in this matter is an extremely important booklet by Stoddard referred to above (15). Implicit in this booklet, in public statements by Alvin C. Eurich, and in the monetary grants made for various projects is that a large share of the solution to the crisis in education must come from automatizing the process through the means of audio-visual devices and techniques.

Basically, the problem has been attacked by scrutinizing the accepted pupil-teacher ratios at all levels and questioning the whole concept of class size. Eurich and Stoddard both insist that, in order to maintain quality of instruction (instruction by able teachers), class size must be increased because the number of able teachers will be smaller. Eurich points out that present pupil-teacher ratios are not sacred. Historically, the idea of one teacher to 25 students, he says, ". . . goes back at least to the middle of the third century. . . . the rule was established by Rabbi Raba . . . '25 students are to be enrolled in one class. If there are from 25 to 40 an assistant must be obtained. Above 40, two teachers are to be engaged' " (3:11). This ruling probably reflects earlier procedures and, of course, existed long before the invention of printing.[5]

Dr. Eurich further makes the point that research has never

[5] It is interesting to note that the idea that a device (in this case, printing) ought to change methodology is not new with Dr. Eurich. Twenty-five years ago, Nicholas Murray Butler said, "The lecture system as a means of communicating facts should have been dispensed with when the art of printing was invented . . ." (10:261)

This content downloaded from
194.169.171.51 on Thu, 16 Oct 2025 05:35:49 UTC
All use subject to https://about.jstor.org/terms

proved conclusively that a small pupil-teacher ratio results in more learning (3:12). This being the case, he and the rather large group of American educators who agree with him feel that class size must be increased in order to rescue teaching quality and properly to reward teaching both economically and socially. For example, O. Meredith Wilson, the president of the University of Oregon, stated:

> But others may ask teachers to accept more students in return for higher salary on the theory that it is better to be 40 feet from distinction than ten feet from mediocrity. As Charles Johnson remarked, ". . . if we are forced to employ mediocre teachers in order to keep our classes small, we may only insure that the teacher communicate his mediocrity in an atmosphere of intimacy." (17:172)

The solutions suggested for increasing class size and the effectiveness of a given teacher include more than mere audio-visual automation of the classroom. The *aide* idea has been given much publicity. This concept is based on the comparison of a teacher's job with that of other professional workers, such as a doctor's. The doctor now has various assistants who relieve him of routine, subprofessional tasks. Studies have shown that much work in the teaching situation is of a subprofessional nature (15:21). Hence, it is felt that teachers can handle larger groups of students for the *professional* phases of their work, allowing aides to carry out routine and clerical functions.[6] The aide program has been widely discussed in the professional and popular literature and, no doubt. most readers are familiar with it. The aide idea might be considered a program separate from that of audio-visual automation except that any consideration of the use of closed-circuit television, for example, presupposes personnel handling the classes at the receiving end. In some of the college experiments, these people have been graduate assistants or other people at the lower instructional levels —aides, really. Further, Mr. Eurich himself has suggested that, part of the time, instruction must be under local people or aides, and makes reference to the fact that "clusters of people and clusters of jobs must be related to clusters of objectives" (3:16).

The aide idea, however, is only an adjunct to the concept of the automatic classroom. The automatization of the educational

[6] It should be emphasized, in this discussion of the efforts to meet the crisis of expanding enrollments and teacher shortage, that the Fund for the Advancement of Education has also spent a great deal of money in efforts to tap heretofore unavailable sources of teachers—liberal arts graduates, housewives who have left teaching, and others—with special training programs which the Fund has subsidized. However, these efforts are not in the main line of the automation movement.

This content downloaded from
194.169.171.51 on Thu, 16 Oct 2025 05:35:49 UTC
All use subject to https://about.jstor.org/terms

process can occur only through the mass use of audio-visual devices and techniques. This was suggested by Dr. Eurich, when he said:

> In addition to making better use of teachers, education must also learn, as it has not yet learned, to make the best possible use of the numerous aids of this kind (audio-visual aids) produced and perfected by man's ingenuity. The phonograph, radio and motion pictures have long been familiar to most American households; but they remain relative strangers in the field of education, where their potentialities are vast. In television, we now have available an almost perfect educational instrument. (4:10) (Material in parentheses supplied.)

The use of television in automatizing the classroom is stated in in the following propositions by Stoddard:

> Let there be experimentation to determine, first, whether there is a practical way to incorporate television into the school program in such manner as to substitute for and lessen what the regular teacher must do so she can do the remainder better and also possibly to teach some things even more efficiently than she could do otherwise; and second, to determine which phases and types of learning experience lend themselves best to the television medium of presentation; and third, whether a high level of teaching efficiency can be attained with fewer trained teachers than would be involved in the usual school organization. (15:43)

When summarized, the thinking represented by Eurich, Stoddard, Faust, and many others associated with various television and film projects may be said to involve:

1. The introduction of mass audio-visual experiences—by picture tape, television or film—into the classroom. This would mean, for example, a whole course in chemistry or French or all of the content of fourth grade social studies taught by master teachers through one or more of these media.

2. Leaving much (perhaps all) of the *systematic* (logical, content, etc.) aspects of teaching to transmission over one or more of these audio-visual media and the *developmental* (personal, social, and growth) aspects to be handled by other persons in the classroom.

3. Large classes as audiences for the transmissions at least part of the school day. (Stoddard proposes a plan for the organization of an elementary school which includes a viewing group in a "resources room" of no less than 150 students while another group of 75 is having some kind of a mass experience in an auditorium. At other periods, students work in small groups (15:45).

4. Developing a group of master teachers who (with assistants) will prepare and deliver the lessons or lectures in a form

This content downloaded from
194.169.171.51 on Thu, 16 Oct 2025 05:35:49 UTC
All use subject to https://about.jstor.org/terms

for audio-visual transmission. These master teachers will draw higher pay equivalent to their ability to operate effectively in the mass media.

5. Carefully planned, highly systematized lessons and lectures equivalent, perhaps, to the careful planning and programming necessary in other forms of automation described in the first paper.

6. Fewer regular classrooms, resulting in a saving of money and time and fewer positions that need to be filled by *trained* teachers, also resulting in a saving of money as well as making a dent in the teacher shortage problem. These savings could be used for the audio-visual equipment (closed-circuit television, etc.) required and to raise the salaries of qualified teachers.

7. Raising the quality of teaching (at least in its systematic aspects) because the master teachers will be experts, both in content and presentation and possess specialized knowledge not available to the line teacher. Thus, no longer will Johnny or Jane have to suffer through general chemistry with the typing teacher because the school had no one qualified to teach chemistry. The chemistry teacher now comes every day or every other day via film or television. Exercises can still be supervised by the typing teacher, but the presentation or content is not her responsibility.

This, then, is the basic theoretical position for automatizing the classroom. There are, in addition, certain factors implicit in this concept that should be stated. These factors have not been directly stated in the literature. They are:

1. A larger number of audio-visual technicians than audio-visual enthusiasts ever imagined in the wildest dreams of post-World War II will have to be found and made available for such a program. Many will be used in preparing materials for the filmed or television presentations; others, presumably, would fit into the "aide" category and render the same service to local teachers.

2. It must be recognized that the classroom automation proposal will attempt to short-cut what might be called normal audio-visual development which, granting Mr. Eurich's point, has really never achieved its potentiality in most schools and colleges of the nation. What, however, will happen to existing audio-visual libraries, and collections of equipment?

This content downloaded from
194.169.171.51 on Thu, 16 Oct 2025 05:35:49 UTC
All use subject to https://about.jstor.org/terms

3. Such an automatization as visualized really has no limits of extension, at least until national boundaries are reached. Dr. Eurich has stated in two places (3:15), (4:13) that some of these television presentations might well be on a *national* basis. Further, the possible tie-ins, through the Educational Television and Radio Center, permit almost indefinite extension of these presentations through television stations and film circulation. In fact, considering the movement as a whole, while most of the activities of the ETRC and other organizations are concerned with *telecasting* as opposed to closed-circuit transmission, the boundaries between the two areas become almost non-existent at times. Stoddard's statement of ETV achievements, for example, jumps from a description of the St. Louis experiment which deals with *broadcasting* to the Hagerstown study, which is *closed-circuit* (15:34). Since the difference between the two is only a technical one concerned with the use or non-use of carrier waves, there is reason to believe that some automatization might originate with broadcasting stations, some with closed-circuit, and some using both at the same time.

4. The entire idea has brought the existing audio-visual movement to a point of serious crisis with no particular evidence that those who have been concerned with the movement over the last two decades are doing or can do much about it. Here and there, as, for example, at San Jose State College in California, the audio-visual people have undertaken the developments in television. For the most part, however, the power of control over this new movement for automatization rests with the organizations described in the early part of this paper.

SUMMARY

This paper has attempted to deal with two aspects of the problem of automatizing the classroom through audio-visual means. These aspects were: (a) the institutional background in which the earlier foundation support of the audio-visual movement by the Payne Fund and the Rockefeller Foundation was compared to the present, much more extensive support of the mass audio-visual approach to teaching supported by the Ford Foundation, its several Funds and the organizations which it underwrites; and (b) the theoretical position requiring audio-visual automation which was shown to be based on the teacher and plant shortage facing this country in the next two decades of swelling school and college

This content downloaded from
194.169.171.51 on Thu, 16 Oct 2025 05:35:49 UTC
All use subject to https://about.jstor.org/terms

enrollments and, therefore, necessitating technological solutions to problems of class size and building space.

For a complete picture, some case histories and research results, as well as a technical analysis of the entire position, would prove useful. It is hoped to supply these three aspects of the study in the next paper, to be followed, if thought desirable, by a fourth discussion which would subject the whole idea to critical scrutiny.

BIBLIOGRAPHY

1. BROOKER, FLOYDE. "DAVI Convention: Market Place of Ideas." *Audio-Visual Instruction* 2:64; February, 1957.
2. CARPENTER, C. R., L. P. GREENHILL, *et al.* *An Investigation of Closed-Circuit Television for Teaching University Courses.* University Park, Pennsylvania: The Pennsylvania State University, 1955.
3. EURICH, ALVIN C. "Better Instruction With Fewer Teachers." *Current Issues in Higher Education, 1956.* Washington, D. C.: Association for Higher Education, a department of the National Education Association, 1956. p. 10-16.
4. EURICH, ALVIN C. "The Teacher's Dilemma." Unpublished paper, June, 1955.
5. FINN, JAMES D. "Automation and Education: I. General Aspects." *Audio-Visual Communications Review* 5:343-59; Winter 1957.
6. *The Ford Foundation Annual Report.* New York: the Foundation, 1956.
7. THE FUND FOR THE ADVANCEMENT OF EDUCATION. *Teachers for Tomorrow.* New York: the Fund, 1955.
8. THE FUND FOR THE ADVANCEMENT OF EDUCATION. *A Report for 1952-1954.* New York: the Fund, 1954.
9. HULL, RICHARD B. "Consider Basic Problems." *AERT Journal* 16:3:5-9; 34-37; December 1956.
10. KNIGHT, EDGAR W. *What College Presidents Say.* Chapel Hill: University of North Carolina Press, 1940.
11. LOMBARD, EDWIN H. "Experimental Teaching by Television." *Proceedings, Western College Association* (Fall meeting, 1956). Fresno, Calif.: the Association, p. 16-26.
12. MCDONALD, RALPH W. "Financing Higher Education: Successful Capital Expansion Programs." *Current Issues in Higher Education, 1956.* Washington, D. C.: Association for Higher Education, a department of the National Education Association. p. 256-262.
13. RESEARCH DIVISION, NATIONAL EDUCATION ASSOCIATION. "Audio-Visual Education in Urban School Districts, 1953-54." *Research Bulletin* 33:3; October 1955.
14. SEMANS, H. H., T. C. HOLY, *et al.* *A Study of the Need for Additional Centers of Public Higher Education in California.* Sacramento: California State Department of Education, 1957.
15. STODDARD, ALEXANDER J. *Schools for Tomorrow: An Educator's Blueprint.* New York: The Fund for the Advancement of Education, 1957.
16. THE TWENTIETH CENTURY FUND. News release by the Fund, September, 1955.
17. WILSON, O. MEREDITH. "The Use of Technical Aids and Assistants." *Current Issues in Higher Education, 1956.* Washington, D. C.: Association for Higher Education, a department of the National Education Association. p. 170-173.

This content downloaded from
194.169.171.51 on Thu, 16 Oct 2025 05:35:49 UTC
All use subject to https://about.jstor.org/terms